George Dyck III

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Research interests: Data systems for digital twins and autonomous machines; federated query processing and data integration; provenance, temporal, and spatio-temporal sensor data; trusted data sharing and data spaces; robotics and agriculture as application domains; computing education.

Education

Ph.D., Biosystems Engineering Conferred December 2025
University of Manitoba
Thesis: *A Novel Digital Twin Framework for Monitoring Inventory and In-Bin Drying in Grain Storage*
Supervisor: Dr. Jitendra Paliwal
Developed a digital twin for stored grain, built as a three-tier web application (Firebase database, Python models, Streamlit front end). Rather than relying on a dense sensor network, the twin used records farmers already keep (mass, moisture, and test weight for each load), stored every load as a timestamped record with full history, and combined that history with continuum-mechanics models to estimate pressure, density, inventory, and drying progress inside the bin and to trace loads from field to market. Pressure models were informed by measurements from a triaxial in-situ sensor I engineered and deployed. The twinned bin was an instrumented research bin at the University of Kentucky, and the work was carried out with industry funding and collaboration from AGCO.

B.Sc., Biosystems Engineering 2020
University of Manitoba
Thesis (CSBE Award, best undergraduate thesis): *Development of Calibration Techniques for the Determination of the Dielectric Properties of Wheat Using a Novel Dielectric Test Cell*

B.A., Mathematics (Minor in Philosophy) 2013
Canadian Mennonite University

Academic Appointments

Postdoctoral Fellow June 2026 – Present
Department of Applied Computer Science, University of Winnipeg
Advisors: Dr. Jitendra Paliwal (University of Winnipeg) and Dr. Adam Rogers (Physics, Brandon University)
Funded by AGCO, directly and through Mitacs
Leads the lab's work on path planning and machine intelligence for autonomous ground robots, with a focus on getting multi-robot control algorithms out of simulation and onto real hardware. Built the lab's robot-fleet testbed and a digital twin of the fleet, which form the experimental base for my proposed data-systems research. Responsible for research design, software development, maintaining the testbed, and mentoring undergraduate researchers.

- Research on path planning and machine intelligence for autonomous robots, including multi-robot formation control and collision behaviour in simulation.
- Built *Faux-GPS*, a ceiling-camera AprilTag localization server for ROS 2 that serves as the lab's robot-fleet testbed and the hardware platform for the Applied Robotics with ROS 2 workshop.
- Built a digital twin of the lab's robot fleet on Faux-GPS, tracking robot positions, maps, and task records.
- Mentoring an undergraduate thesis student working on GPS localization for rovers, and serving as faculty advisor to the University of Winnipeg Robotics Club.

Academic Teaching Appointments

Course Developer & Instructor 2025 – Present
Department of Applied Computer Science / Department of Physics, University of Winnipeg
Physical Computing (PHYS-2803 / ACS-2803), cross-listed Physics and Applied Computer Science
Asked to build this course from nothing. Wrote the syllabus, lectures, labs, assignments, and exams. The course takes students who may never have touched a breadboard to the point where they can design and program an autonomous vehicle.

- Designed the full curriculum from scratch: circuit fundamentals, digital logic, and embedded programming in C++ on Arduino.
- Built 10 hands-on labs and 10 technical assignments, culminating in a capstone in which students engineer and program autonomous obstacle-avoiding vehicles.

Applied Robotics with ROS 2 workshop (36 hours over four weeks; not yet a credit course), funded by the AGCO Foundation ($27,000)
Designed, built, and taught this workshop as a pilot for a future credit course, including all ten labs, the hardware and software infrastructure, and the public capstone event.

- Designed a project-based ROS 2 curriculum for diverse majors, built backwards from farm tasks: ten labs from simulation to perception to real hardware, ending in a public "AgTech Triathlon" capstone (seeding, spraying, avoidance, and safety) on real robots.
- 21 participants (17 in person, 4 remote) from physics, computer science, and other majors; exit survey rated the workshop 4.7/5 (13 student responses).
- Built the course infrastructure: public course website (GitHub Pages), reproducible Docker development containers, and a ceiling-camera AprilTag localization system (Faux-GPS) that serves as live GPS for student robots in the capstone arena.
- Replaced formal lab reports with weekly reflective dossiers; offered credit through PLAR to support non-traditional learners.

Instructor 2023 – Present
Department of Mathematics, Canadian Mennonite University
Calculus I (MATH-1020) · *Calculus II* (MATH-1030) · *Ordinary Differential Equations* (MATH-2040)
Sole instructor, responsible for every part of course delivery, from the syllabus and lectures to setting and marking all assessments. Classes are small and mixed, so they run as flipped classrooms where class time is spent working problems together.

- Teach small flipped classrooms of students from a wide range of academic backgrounds.
- Developed all course materials: syllabi, lectures, assignments, and exams.

Teaching Assistant 2022 – 2023
Department of Biosystems Engineering, University of Manitoba
Design of Structural Components in Machines (BIOE-3400) · *Instrumentation and Measurement for Biosystems* (BIOE-3270)

- Led weekly tutorials on engineering mathematics and analysis software; developed supplementary assignments and marking schemes.

Publications

Doctoral dissertation

- **Dyck, G.** (2025). *A Novel Digital Twin Framework for Monitoring Inventory and In-Bin Drying in Grain Storage*. Ph.D. dissertation, Department of Biosystems Engineering, University of Manitoba. Available on MSpace

Peer-reviewed journal articles

[1] **Dyck, G.**, Rogers, A., Montross, M., Hildebrand, K., Turner, A., Paliwal, J., Barry, F., & Jarro, C. (2025). A triaxial in-situ stress measurement system for investigating radial variability of the vertical-to-lateral pressure ratio in grain silos. *Computers and Electronics in Agriculture*. doi:10.1016/j.compag.2025.110698

[2] **Dyck, G.**, Rogers, A., & Paliwal, J. (2024). A review of analytical methods for calculating static pressures in bulk solids storage structures. *KONA Powder and Particle Journal*, 41, 108–122. doi:10.14356/kona.2024013

[3] Rogers, A., **Dyck, G.**, Paliwal, J., Hildebrand, K., Montross, M. D., & Turner, A. P. (2024). The Janssen effect and the Chini ordinary differential equation. *Powder Technology*, 436, 119493. doi:10.1016/j.powtec.2024.119493

[4] Rogers, A., **Dyck, G.**, Zhang, Q., Hildebrand, K., & Paliwal, J. (2024). A general continuum modelling approach for variable shear stress as a separable function in stored bulk solids. *Powder Technology*, 434, 119331. doi:10.1016/j.powtec.2023.119331

[5] **Dyck, G.**, Hawley, E., Hildebrand, K., & Paliwal, J. (2023). Digital twins: A novel traceability concept for post-harvest handling. *Smart Agricultural Technology*, 3, 100079. *(75 citations)* doi:10.1016/j.atech.2022.100079

In preparation

- **Dyck, G.**, Rogers, A., Hawley, E., & Paliwal, J. A control scheme for the motion of a swarm of robots I: Theory and numerical approach. *Manuscript in preparation.*
- **Dyck, G.**, Rogers, A., Hawley, E., Paliwal, J., Bidinosti, C., & Beck, M. A control scheme for turn-constrained robot swarms II: Faux-GPS and a shared-controller framework for sim-to-real evaluation. *Working draft.*
- **Dyck, G.**, et al. A digital twin tool for post-harvest grain inventory management.
- **Dyck, G.**, et al. Empirical fitting functions for failure criteria.

Patents

- Triaxial in-situ pressure sensor for grain storage. *Information will be provided upon request.*

Conference Presentations and Invited Talks

- 1st Manitoba Digital Agriculture Symposium, University of Manitoba, Winnipeg (2026), poster. *Teaching ROS 2 Through Farmer-Focused Robotics Projects: A Practical, Project-Based ROS 2 Curriculum for Diverse Majors Building Autonomous Agricultural Robots.* **Dyck, G.**, Bidinosti, C. P., Beck, M. A., Hawley, E., Rogers, A., & Paliwal, J.
- CSBE Annual Conference, Winnipeg (2024). *Triaxial Pressure Sensor: Enhancing Grain Storage Management.*
- CSBE Annual Conference, Lethbridge (2023). *Conceptualizing a Digital Twin for Grain Bins: Mathematical Models for Pressure and Density Changes.*
- AGCO Grain Symposium (2022), invited. *Digital Twins & Food Traceability: Industry Optimization and Consumer Choice.*
- CSBE Annual Conference, PEI (2022). *Eliminating Food from Nowhere: Digital Twins and the Supply Chain.*
- Department of Biosystems Engineering Seminar, University of Manitoba (2022). *Digital Twins and the Future of Food*; *Janssen Packing Equations: A Historical Perspective.*
- AGCO Voice of Customer (2021), invited. *Digital Twins.*
- INGENIUM (2020). *UM agBOT: Autonomous Farming Vehicle.*

Research Funding, Fellowships, and Awards

Research funding and fellowships

AGCO postdoctoral research funding, direct and through Mitacs ($70,000 per year)	2026 – Present
AGCO Foundation, *Applied Robotics with ROS 2* workshop ($27,000)	2026
Mitacs Accelerate, industry-partnered research funding with AGCO	2020 – 2024
University of Manitoba Graduate Fellowship (UMGF), Ph.D.	2022 – 2024
Stewart Pugh External Study Scholarship	2023 – 2024
Lord Selkirk Association of Rupert's Land Agriculture Scholarship	2023 – 2024
University of Manitoba Graduate Fellowship (UMGF), Master's	2021 – 2022

Awards and distinctions

CSBE Undergraduate Thesis Award (best undergraduate thesis)	2020
3rd Place, International Autonomous Agricultural Robotics (agBOT) Challenge, Purdue University	2019
CSBE Undergraduate Design Award	2019
John Shewchuk Engineering Design Award	2019
Undergraduate Research Award (URA); UMSU Undergraduate Research Award	2019
NSERC Undergraduate Student Research Award	2018
Faculty of Engineering Centenary Scholarship; Dr. Gerry Price Scholarship	2018
Dr. G.E. Laliberte Scholarship	2017
Dean's Honour List, Faculty of Engineering	2014 – 2020

Research Experience

Doctoral Researcher & Co-Founder, Grain Packing Research Collective 2021 – Present
University of Manitoba; international collaboration
Beyond my own thesis research, co-founded and coordinate a working group of researchers at the University of Manitoba, Brandon University, the University of Kentucky, Clemson University, and AGCO who study pressure and packing in stored grain; lead or co-author the group's papers.

- Developed the concept of a digital twin for grain storage: mathematical models and data pipelines that estimate in-bin pressure and density without dense sensor networks.
- Engineered and deployed a novel triaxial in-situ pressure sensor (Arduino, microstrain sensors, I^2C), since patented, with results published in *Computers and Electronics in Agriculture*.

Undergraduate Research Assistant (URA) 2019
Department of Biosystems Engineering, University of Manitoba

- Studied how camera orientation on an autonomous farm vehicle affects a remote supervisor's situational awareness; designed the methodology and built the sensor system and camera mounts.

NSERC Undergraduate Research Assistant 2018
Department of Biosystems Engineering, University of Manitoba

- Produced biodegradable microbeads from bacterial PHAs; analyzed data in R; presented at the CSBE conference.

Industry Experience

Student Programmer (Full-Stack Developer) 2019
Integrated Application Services, Agriculture and Agri-Food Canada (Federal Government)
Member of a federal software development team building internal web applications used by AAFC staff. Responsible for features end to end, from SQL against the Oracle database, through server-side logic, to the user interface.

- Built internal single-page web applications on an Oracle database backend: wrote SQL queries and server-side code in Groovy on Grails (JVM), with Vue.js, jQuery, and Bootstrap front ends.
- Worked with clients and business analysts to gather requirements; used Git and JIRA for version control and issue tracking; took part in reciprocal code review.
- Subsequently offered a full-stack developer position with a federal bioinformatics team (Ottawa); declined to pursue graduate research.

Engineering Intern 2020
AGCO Corporation

- Built a database-backed tool (create, read, update, delete) for managing experimental records.
- Debugged and improved production algorithms; calibrated grain moisture measurement technology and analyzed data in MATLAB; wrote standard operating procedures.

Database and Web Applications for Community Organizations

Data-backed web applications designed and built for community groups and events in Winnipeg, often as a volunteer. These are small but complete systems: data model, back end, real-time or transactional logic, front end, deployment, and support on the day they go live.

Full-Stack Developer (volunteer), Bike, Walk, Run or Roll-a-thon 2026
Community fundraiser, Winnipeg
A friend needed to raise money quickly to travel to Spain for emergency surgery. Volunteered as the sole full-stack developer, designing and building the fundraiser's entire online system, front end and back end, on Firebase. Participants could register for the event; supporters could pledge and donate to individual participants; and organizers could track donations and totals as they came in. Designed the data model for participants, pledges, and donations, and automated emails so supporters received confirmations and follow-ups without manual work

from organizers. The system supported dozens of participants and handled donations totalling tens of thousands of dollars.

Developer, *Big Sun Wall* interactive projection 2026
Big Sun festival, Winnipeg
A crowd-sourced digital graffiti wall for the Big Sun festival. Attendees scanned a QR code on posters around the venue, which opened a retro "Paint"-style drawing app on their phone; they painted a tag and submitted it, and after moderation it appeared on a projection above the bar about two and a half storeys tall (roughly 7.5 m $\times$ 13 m from a single 1080p projector). It used a Node.js server with real-time WebSocket (Socket.IO) messaging, storage of submitted images, a moderation queue, a placement algorithm that spreads pieces across the wall so it fills evenly over a night, a password-protected moderator view, and the mobile drawing client. Deployed it to a cloud host, wrote an automated end-to-end test suite and operating instructions for venue staff, and ran moderation on the night.

Mentorship and Student Supervision

Undergraduate Thesis Mentor 2026 – Present
University of Winnipeg
Mentoring, informally and at the request of Dr. Michael Beck, an Applied Computer Science student on a ROS 2 thesis project on GPS localization for rovers. Formal meetings about once a month, plus frequent informal discussion since the student is also in my course. Helped define the project scope, guide the technical work and writing, and supported the student in presenting the work as a poster at the Canadian Black Scientists Network (CBSN) conference in 2026.

Faculty Advisor 2026 – Present
University of Winnipeg Robotics Club
Faculty contact and advisor for the student robotics club. Helped organize the club's first meeting, worked with members to set clear goals for moving forward, and share advice from founding and captaining the UM-agBOT team.

Co-Founder & Captain; Volunteer Advisor 2018 – Present
University of Manitoba Agricultural Robotics Club (UM-agBOT)
Co-founded the team and served as captain. Recruited and led three to eight undergraduate students at any given time, set the technical direction, and secured industry sponsorship to fund the build. The team designed and built an autonomous weed-detecting and spot-spraying robot (Python, ROS, distributed embedded systems) and placed 3rd at the 2019 international agBOT Challenge at Purdue University. Continue to volunteer with the team since 2021, giving design reviews and lecturing members on Arduino programming.

Mentor, "Superhero Science" STEM Outreach 2020 – 2025
Rural Manitoba high schools
High school students were each assigned a chemical element and invented a superhero based on its properties. Reviewed students' ideas and responded with recorded videos: validating their science, suggesting ways to develop their heroes further, connecting their elements to my research, and answering their STEM questions. Mentored in two rounds of the program.

Industry Partnerships

- **AGCO Corporation**: former engineering intern (2020); industry partner on Mitacs-funded doctoral research (2020–2024); co-author on digital twin research (2023); funds current postdoctoral position (2026–present); invited speaker at AGCO events (2021, 2022); the AGCO Foundation funded the University of Winnipeg ROS 2 workshop ($27,000; 2026).

Creative Technology and Community Engagement

Alongside research and teaching, active as a digital artist, designer, musician, and creative technologist in Winnipeg's music and arts community under the name V01D R4ST3R. All of this work is offered on a sliding scale so that local artists, venues, and community members can afford it. It uses the same skills I teach (programming, 3D graphics,

real-time systems, and hardware) in public-facing projects, and connects me to a wide network of artists, venues, and audiences in the city.

Interactive installations and games

Game Developer, *Glitch Lich Revolution* Sept. 2025

Urban Shaman Contemporary Aboriginal Art; HYPER ART, Nuit Blanche Winnipeg, Théâtre Cercle Molière

Sole software developer for a collaborative live performance presented by Urban Shaman during Nuit Blanche. Designed and programmed a *Dance Dance Revolution*-style rhythm game in Unity (C#) in which four audience members at a time played on dance-pad controllers, shown on a large on-stage screen during DEAFWIZRD's live breakcore set; built the 3D assets in Blender. The game's story, an anti-colonial fable in which Two-Spirit defenders draw on ancestral languages, songs, and dances to storm the forts of an undead cyborg Queen Victoria, let players take part in the work's message by dancing. With Phoenix Hourie (art) and Carter Lowe / DEAFWIZRD (music); curated by Justin Bear L'Arrivee.

3D Artist & AR Developer, *IMABOX* 2026

Market Square, Nuit Blanche Winnipeg (central exhibit); with artist Bîstyek

Collaboration with Bîstyek, a Winnipeg painter of Kurdish background who came to Canada from Syria and whose work, featured at the Winnipeg Art Gallery (WAG), addresses displacement and resilience. The project turned his paintings into an animated 3D piece that viewers open on their own phone through a QR code and see in augmented reality on Android or iOS, with no app to install. Built the 3D models from his paintings, rigged them and painted the skin weights, cleaned up the animation (animated by another collaborator), rendered material for a promotional video, and built the web-AR experience. Shown at Market Square as one of the central exhibits of Nuit Blanche Winnipeg 2026. Bîstyek has asked to continue collaborating.

Projection Systems Developer, *Hocketus* 2026

Savant Flaneur album release, Winnipeg

For Winnipeg avant-garde chamber collective Savant Flaneur's album release, designed and wrote a real-time projection system in Python that takes two musicians' audio live from an audio interface and visualizes their interplay (who leads, who yields, when they interrupt or sound together) rather than the spectrum, with track changes cued over OSC and MIDI. The project began my work in real-time audio signal processing, which continues with the goal of building my own audio plugins.

Game Developer, German folklore video game 2026 – Present

In collaboration with composer Radagast

Co-developing a video game with international composer Radagast that tells stories from German folklore. Responsible for the 3D art, programming, and game design; Radagast writes the music and story. A stylized first-person adventure built from real imagery of German landscapes and folk culture, focused on puzzles and decision-based storytelling rather than combat. Follows three earlier animations made for his music (see below).

Developer & Game Designer, Luana Moth alternate reality game 2025

luanamoth.com

Commissioned by musician Luana Moth to build an alternate reality game (ARG) for the artist's website, including an "in-universe" browser game (*Project Quassia: Invitation*) written in Phaser.js with 2D pixel art and lore puzzles. Designed and programmed the whole project.

Interactive Web Developer 2024 – Present

Personal and commissioned projects, georgedyck.com

Custom interactive websites for artists and for my own work, including a portfolio built as a working simulated Windows 95-style desktop operating system ("GeorgeOS") with movable windows, a Start menu, and hidden puzzles; a command-line alternate reality game for the band MOYAMOYA (moyamoya.ca); and an interactive text artwork on the Book of Job.

Music videos and animation

Director & 3D Animator, Tired Cossack – "Cab" 2026

Music video, from the album ZIMA

Full-length official music video for Winnipeg band Tired Cossack's song "Cab." Directed it and built and animated every scene in Blender. Released on YouTube in August 2026, the video had 78,961 views by October 2026.

3D Animator, Nadarak – "Fixie" 2026

Music video

Made the music video for British Columbia artist Nadarak's song "Fixie," styled as a 1990s PlayStation-era racing game. All modelling, animation, and rendering in Blender.

Animator, collaborations with composer Radagast 2026

International collaboration

Created three animations set to Radagast's original music, and designed the poster for his Winnipeg show at Public Domain (presented by Eye and Ear Control Records and Send & Receive).

3D Animator, venue and business visualizers 2025 – Present

Winnipeg businesses and venues

Commissioned and collaborative 3D animations for local businesses and venues, in a low-poly PlayStation aesthetic: a bowling animation for Park Alley punning on the "STRIKE" of the 1919 Winnipeg General Strike; an animation turning the Hoagie Boys storefront into a hoagie; a visualizer for local record distributor Eye and Ear Control Records; a looping GameCube-style animation for Public Domain's recurring "PBS" events; and a 3D animated version of an original character by local artist Basement Lagoon (Austin Boulton) for Fantasy Club Books.

Graphic design and type design

Type Designer, custom typefaces 2025 – 2026

Loewen Guitars (luthier); Dorothy Wolfe (fashion designer)

Commissioned by two Winnipeg businesses to design custom typefaces for their brands. Each translates historical needlework patterns into a complete digital font, treating the stitch grids of old patterns as an early form of pixel art.

Poster and Logo Designer 2024 – Present

Winnipeg music venues, artists, and businesses

Concert posters printed and posted around the city and shared online, usually built from my own 3D renders. Recent posters include shows at the West End Cultural Centre, Sidestage, The Handsome Daughter, and Public Domain. Also designed logos for DJ Glitch Witch and for Hardcore Painting, a local painting company.

Band and event production

Bassist, Co-songwriter, Art Director, and Manager, MOYAMOYA 2024 – Present

Winnipeg band

Bassist and co-songwriter; the band writes its music collaboratively. Also compose and produce the interludes in our set, and recorded and produced one of our songs. Handle much of the band's business and creative work: organizing and promoting shows (booking lineups, designing and putting up posters, selling tickets); album artwork and visualizers for single releases; 3D animations for the live show; and all merchandise design (two sold-out print runs), along with merchandise sales and managing the band's bank account and finances.

Composer and Producer Ongoing

Independent

Write classically inspired music and produce jungle, breakcore, and drum and bass in Ableton Live and Reaper, with several recording projects under way. Play classical guitar, guitar, and bass; take composition lessons, with a particular interest in counterpoint and music for video games.

Technical Skills

Data & databases	SQL, Oracle, Firebase, relational schema design, data acquisition, statistical analysis (R, MATLAB)
Programming	Python, C++, C#, JavaScript (Node.js, Socket.IO, Vue.js, jQuery, Phaser), Groovy on Grails, MATLAB, R
Robotics & systems	ROS / ROS 2, Docker, Linux, Git, JIRA; Arduino, Raspberry Pi, NVIDIA Jetson; I^2C, PWM, sensor integration
Creative & design	Unity, Blender (modelling, rigging, animation), Adobe After Effects, Photoshop, type design, SolidWorks, AutoCAD, LaTeX
Audio	Ableton Live, Reaper, recording and production; real-time audio processing (Python), OSC/MIDI
Pedagogy	Flipped classroom, project-based learning, curriculum design, lab development

Media Coverage

UWinnipeg News: "Field wheels: exploring agricultural robotics" (2026) · UM Today · Resource Magazine · CSBE Perspectives · JCA Technologies · LINAK podcast